

Abstract No.

Liquid Biopsy-Based Methylation and Fragment Signatures in Intracranial Meningiomas

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Intracranial meningiomas are the most common primary tumors of the central nervous system. Although the majority are histologically benign, their biological behavior is highly heterogeneous, with some tumors demonstrating aggressive growth, recurrence, or resistance to treatment. Gamma Knife stereotactic radiosurgery (GKRS) is widely used for the management of selected meningiomas. However, predicting individual treatment response remains challenging because reliable non-invasive molecular biomarkers are lacking. Circulating cell-free DNA (cfDNA) represents a promising source of tumor-derived molecular information, with both DNA methylation patterns and fragment profiles reflecting disease-specific biological processes. This study aims to characterize plasma cfDNA methylation and nucleosome fragment populations in patients with intracranial meningiomas and healthy individuals to identify biomarkers associated with tumor presence and response to GKRS. The study cohort comprised 53 patients with intracranial meningiomas who underwent blood sampling immediately before GKRS and at a six-month follow-up, together with 20 age-matched healthy controls. Plasma cfDNA was isolated using the QIAamp Circulating Nucleic Acid Kit and quantified using the Qubit fluorometric assay. Fragment size distribution was assessed using the Agilent TapeStation system, enabling characterization of mono-, di-, and tri-nucleosome cfDNA populations. Genome-wide profiling of 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC) was performed using Oxford Nanopore sequencing. Fragment profiles and methylation signatures were compared between patients and controls and evaluated in relation to clinical characteristics. The ongoing study identified distinct methylation and fragment signatures associated with meningioma profiling and patient response to GKRS.



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